

# CaptionAI: personalized and network assistant

## Project Description:

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The **Personalized Networking Assistant** is an AI-powered web application designed to help users build meaningful professional connections by generating personalized conversation starters based on event topics and user interests. The platform leverages **DistilBERT** to extract key themes from event descriptions and **GPT-2** to generate context-aware conversation suggestions.

The application includes a **Theme Extraction Module**, which identifies important keywords and topics from networking event descriptions, a **Conversation Starter Generator**, which provides multiple personalized icebreakers, a **Fact-Checking Module**, which verifies generated information using the Wikipedia API, and a **Conversation History & Feedback Module**, which stores user interactions and feedback to improve future recommendations.

The project is built using **FastAPI** for the backend, **Streamlit** for the frontend, and **Pytest** for unit testing. It provides a responsive, user-friendly interface with modular architecture, making networking preparation faster, smarter, and more effective.

## Scenarios

### Scenario 1:

A user is attending an **AI for Sustainable Cities** conference and enters the event description along with their interest in **Artificial Intelligence**. The system extracts key themes using **DistilBERT** and generates three personalized conversation starters using **GPT-2**. It also verifies key facts using the Wikipedia API and presents engaging icebreakers to help the user confidently start conversations with other attendees.

### Scenario 2:

A user plans to attend a **Cloud Computing and Cybersecurity Meetup** and provides the event details. The Personalized Networking Assistant identifies topics such as cloud security, data privacy, and DevOps, then generates context-aware conversation starters tailored to professionals in the technology industry. The user can save the generated suggestions and review them before the event.

### Scenario 3:

A user is preparing for a **Startup Innovation Summit** and wants to network with entrepreneurs and investors. The system analyzes the event description, extracts themes related to startups, innovation, and investment, and generates personalized conversation prompts. The application also stores the user's conversation history and feedback, enabling improved recommendations for future networking events.

backend orchestrates data validation and prompt construction. It interfaces with the Groq API using the LLaMA 3.3-70B model to generate tailored social media content, with secure deployment managed locally via Flask. Description: CaptionAI utilizes a modular three-tier architecture to deliver rapid, AI-driven content generation. The frontend provides a responsive user interface built with HTML and CSS while the Flask-based and exposed publicly through Ngrok. (Note: If you are using

database in your project definitely you need to add ER Diagram along with description) Pre-requisites: ● **Flask Framework Knowledge:** Flask Documentation ● **Groq API Familiarity:** Groq API Documentation ● **HTML, CSS, and JavaScript Skills:** W3Schools HTML/CSS/JavaScript Tutorials ● **Python Programming Proficiency:** Python Documentation ● **Version Control with Git:** Git Documentation ● **Development Environment Setup:** Flask Installation Guide ● **Ngrok for Public Deployment:** Ngrok Documentation

Here's the same section rewritten for your **Personalized Networking Assistant** project.

### Description

The **Personalized Networking Assistant** utilizes a modular three-tier architecture to deliver intelligent, AI-powered networking assistance. The frontend is built using **Streamlit**, providing a responsive and interactive user interface where users can enter event details and networking interests. The backend is developed using **FastAPI**, which handles input validation, theme extraction, conversation generation, fact verification, and user feedback management.

The application integrates **DistilBERT** to extract key themes from event descriptions, **GPT-2** to generate personalized conversation starters, and the **Wikipedia API** to verify factual information. User conversation history and feedback are stored to improve future recommendations. The solution follows a modular architecture, making it scalable, maintainable, and easy to extend with additional AI models or APIs.

**ER Diagram:** Not Applicable (No Database Used).

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### Pre-requisites

- **FastAPI Framework Knowledge:** FastAPI Documentation
- **Streamlit Knowledge:** Streamlit Documentation
- **Python Programming Proficiency:** Python Documentation
- **Natural Language Processing (NLP):** DistilBERT and GPT-2 Basics
- **Wikipedia API Integration:** Wikipedia API Documentation
- **Unit Testing with Pytest:** Pytest Documentation
- **Version Control with Git:** Git Documentation
- **Development Environment Setup:** Python Virtual Environment (venv) and pip
- **Visual Studio Code:** For development and debugging

This version matches your **Personalized Networking Assistant** project and is suitable for your SmartBridge documentation.

For your **Personalized Networking Assistant** project, you can replace it with the following:

### Activity 1.2: Research and Select the Appropriate AI Models

The following aspects were researched to select the most suitable AI models for the Personalized Networking Assistant:

- **Understand the Project Requirements:** Review the application's objectives, including extracting key themes from networking event descriptions, generating personalized conversation starters, verifying factual information, and improving networking experiences.
- **Explore NLP Model Documentation:** Study the documentation of Hugging Face Transformer models, including **DistilBERT** for text classification and theme extraction, and **GPT-2** for natural language generation. Also review the **Wikipedia API** for fact verification.

- **Evaluate Model Performance:** Compare different models based on accuracy, inference speed, computational efficiency, and the quality of generated conversation starters. DistilBERT was chosen for its lightweight architecture and fast theme extraction, while GPT-2 was selected for generating coherent and context-aware conversation prompts.
- **Select the Optimal Models:** Choose **DistilBERT** for extracting meaningful themes from event descriptions and **GPT-2** for generating personalized networking conversation starters. Integrate the **Wikipedia API** to verify factual information, ensuring reliable and trustworthy suggestions for users.

Activity 1.3: Define the Architecture of the Application • **Draft an Architectural Diagram:** Create a visual representation of the application architecture, including components like the frontend (HTML, CSS, JavaScript), Flask backend, and Groq API integration. • **Detail Frontend Functionality:** Outline how users interact with the application entering product/campaign information, selecting a platform (Instagram or LinkedIn), and choosing a tone (Professional, Casual, or Promotional). • **Outline Backend Responsibilities:** Specify how the Flask backend handles incoming POST requests to /generate, validates and sanitizes user input, builds a structured prompt, and communicates with the Groq API.

For your **Personalized Networking Assistant** project, write the following:

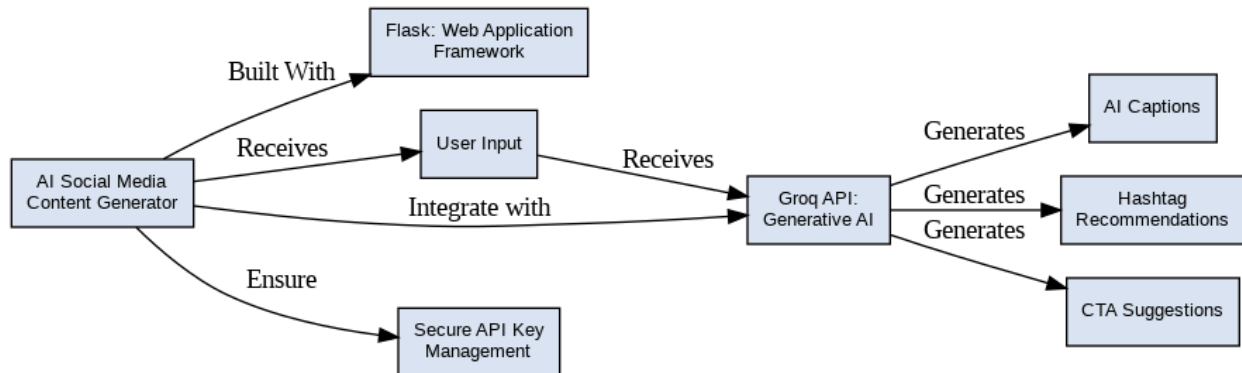
### Activity 1.3: Define the Architecture of the Application

- **Draft an Architectural Diagram:** Create a visual representation of the application architecture, including the **Streamlit frontend**, **FastAPI backend**, **DistilBERT** for theme extraction, **GPT-2** for conversation generation, and the **Wikipedia API** for fact verification.
  - **Detail Frontend Functionality:** Describe how users interact with the application by entering a networking event description, selecting their interests or networking goals, and submitting the request to generate personalized conversation starters. The frontend also displays extracted themes, fact-check results, conversation history, and allows users to provide feedback.
  - **Outline Backend Responsibilities:** Explain how the **FastAPI backend** receives user requests, validates and processes the event description, extracts key themes using **DistilBERT**, generates personalized conversation starters using **GPT-2**, verifies information through the **Wikipedia API**, manages conversation history and feedback, and returns the generated results to the Streamlit frontend.
- **Describe AI Integration Points:** Define how the backend constructs prompts using TONE\_DESCRIPTIONS and PLATFORM\_TIPS dictionaries, calls the Groq API, parses the JSON response via `extract_json()`, and returns structured results to the frontend.

For your **Personalized Networking Assistant** project, replace that point with:

- **Describe AI Integration Points:** Define how the **FastAPI backend** receives the user's networking event description, preprocesses and validates the input, extracts key themes using **DistilBERT**, generates personalized conversation starters using **GPT-2**, verifies important facts through the **Wikipedia API**, and returns the generated conversation starters, extracted themes, fact-check results, conversation history, and feedback to the **Streamlit frontend** in a structured format.

## Technical Architecture:



**Description:** CaptionAI utilizes a modular three-tier architecture to deliver rapid, AI-driven content generation. The frontend provides a responsive user interface built with HTML and CSS while the Flask-based backend orchestrates data validation and prompt construction. It interfaces with the Groq API using the LLaMA 3.3-70B model to generate tailored social media content, with secure deployment managed locally via Flask and exposed publicly through Ngrok.

**(Note: If you are using database in your project definitely you need to add ER Diagram along with description)**

## Pre-requisites:

- **Flask Framework Knowledge:** [Flask Documentation](#)
- **Groq API Familiarity:** [Groq API Documentation](#)
- **HTML, CSS, and JavaScript Skills:** [W3Schools HTML/CSS/JavaScript Tutorials](#)
- **Python Programming Proficiency:** [Python Documentation](#)
- **Version Control with Git:** [Git Documentation](#)
- **Development Environment Setup:** [Flask Installation Guide](#)
- **Ngrok for Public Deployment:** [Ngrok Documentation](#)

## Project Workflow:

### Milestone 1: Model Selection and Architecture

- **Activity 1.1:** Generate a Groq API key and configure it securely in the project environment.
- **Activity 1.2:** Research and select the appropriate generative AI model from Groq for social media content generation.
- **Activity 1.3:** Define the architecture of the application, detailing interactions between the frontend, backend, and AI integration.
- **Activity 1.4:** Set up the development environment, installing necessary libraries and dependencies for Flask and the Groq API.

### Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop the core functionalities: Caption Generator, Hashtag Suggester, and Call-to-Action Generator.
- **Activity 2.2:** Implement the Flask backend to manage routing and user input processing, ensuring smooth API interactions.

### Milestone 3: App.py Development

- **Activity 3.1:** Write the main application logic in app.py, establishing routes for each feature and integrating AI responses.

### Milestone 4: Frontend Development

- **Activity 4.1:** Design and develop the user interface using HTML, CSS, and JavaScript, ensuring a responsive and intuitive layout.
- **Activity 4.2:** Create dynamic templates with Flask's render\_template to display results based on user interactions.

### Milestone 5: Deployment

- **Activity 5.1:** Prepare the application for local deployment by configuring the server environment and ensuring all dependencies are correctly installed.
- **Activity 5.2:** Deploy the application publicly using Ngrok to expose the local Flask server over a secure public URL.

### Milestone 6: Conclusion

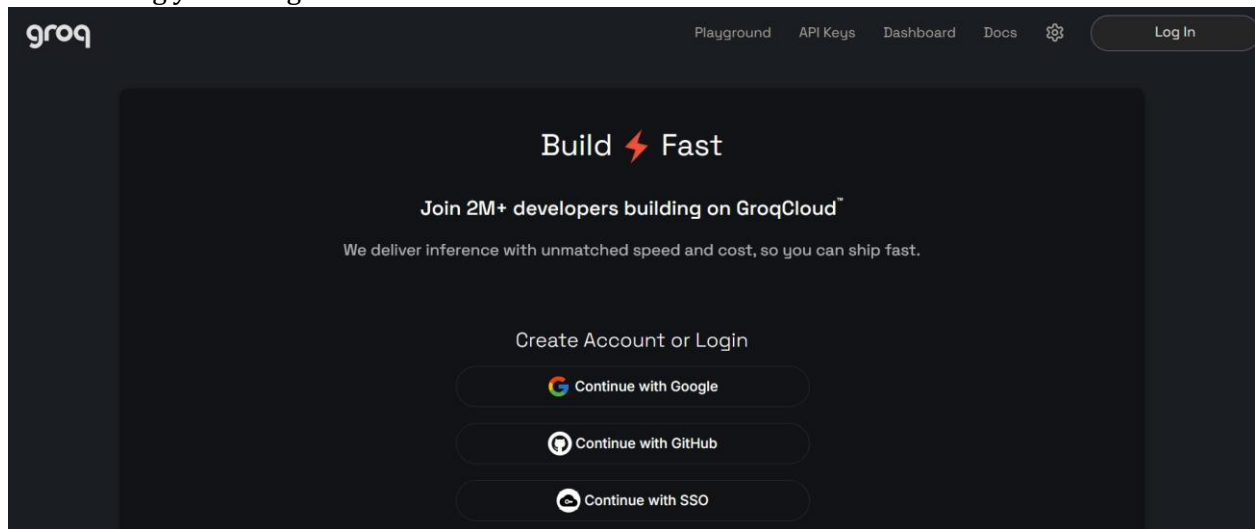
## Milestone 1: Model Selection and Architecture

In this milestone, we focus on selecting the appropriate generative AI model from Groq for our social media content generation needs. This involves researching the capabilities and performance of various models that Groq offers, ensuring that the chosen model aligns well with the application's objectives of generating captions, hashtags, and CTAs across different platforms and tones.

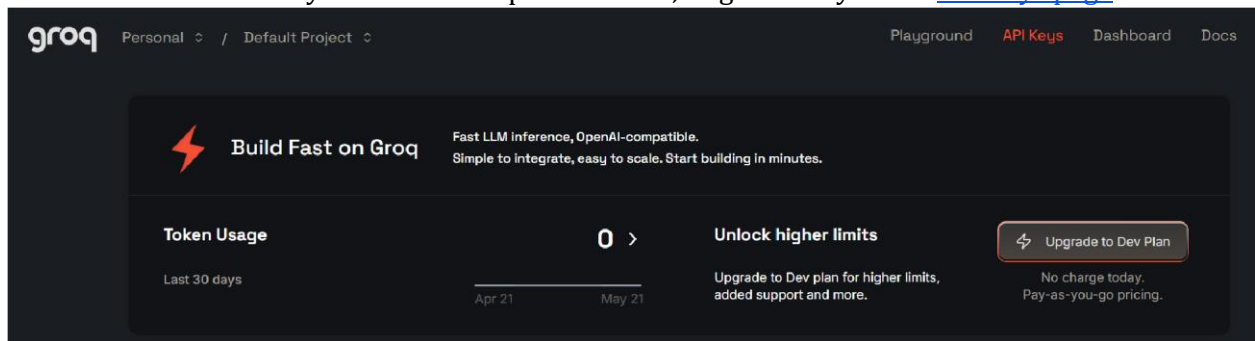
### Activity 1.1: Generate a Groq API Key

Before the application can communicate with the Groq LLM, you need a valid Groq API key. Follow the steps below to create one and connect it securely to the project.

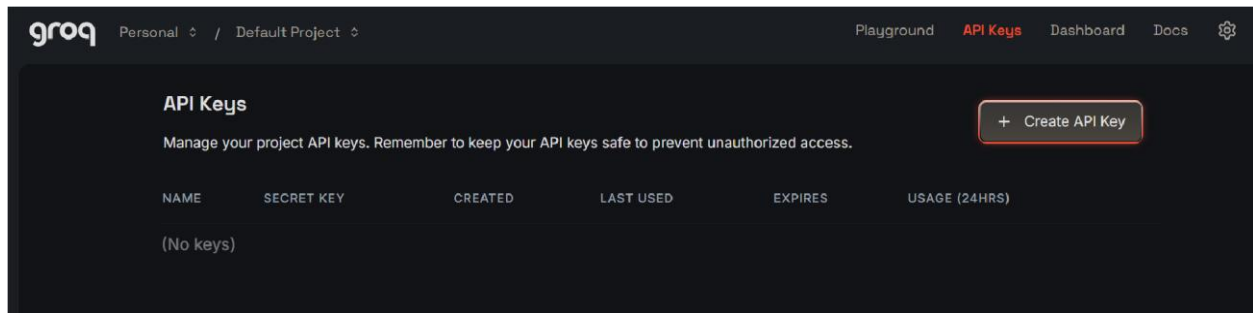
- **Step 1 — Create a Groq Account:** Visit <https://console.groq.com> and sign up for a free account using your Google account or email address.



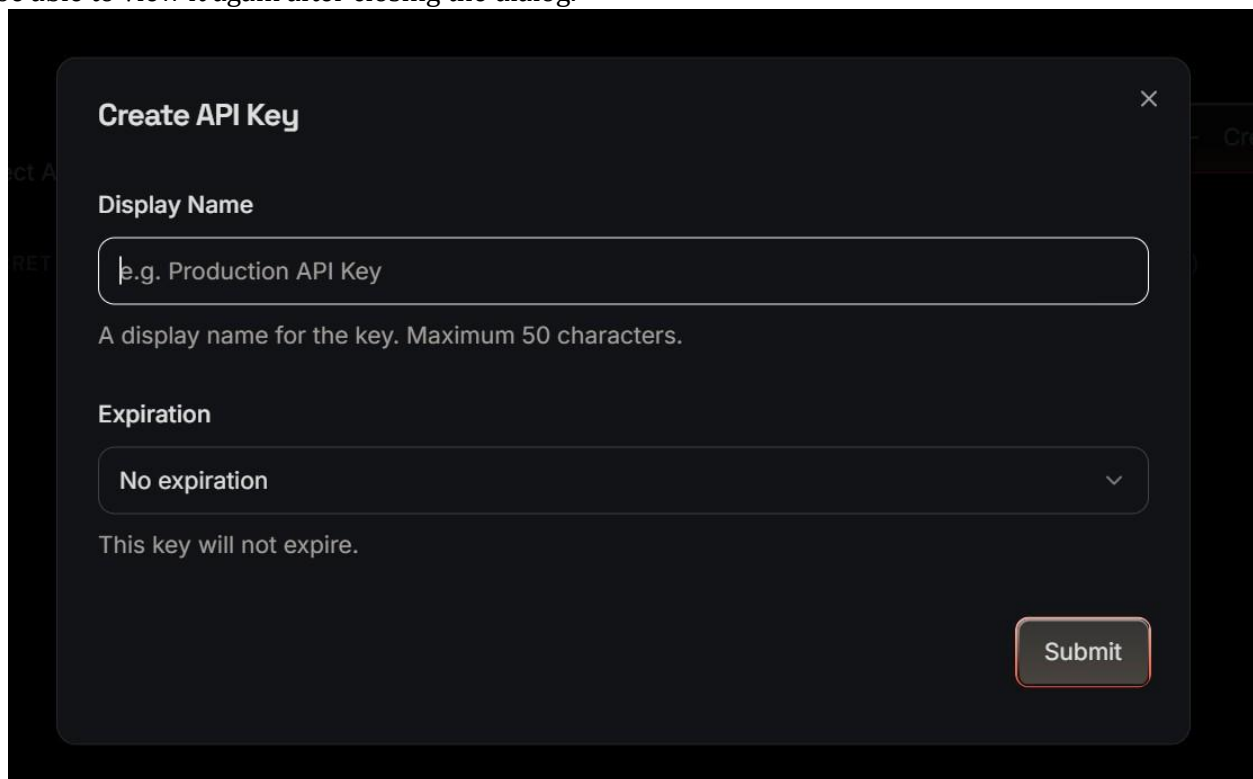
- **Step 2 — Navigate to API Keys:** After logging in, click on your profile icon in the top-right corner and select “API Keys” from the dropdown menu, or go directly to the [API Keys page](#).



- **Step 3 — Create a New API Key:** Click the “Create API Key” button. Give your key a descriptive name (e.g., captionai-key) so it is easy to identify later.



- **Step 4 — API Key:** Now give a “Display Name” and “Expiration” is optional, and Click on “Submit”. The you will get your API key, Copy it immediately and store it in a safe place you will not be able to view it again after closing the dialog.



- **Step 5 — Add the Key to Your Project:** Create a .env file in the root of your project directory (if it does not already exist) and paste the key as shown below:  
GROQ\_API\_KEY=your\_copied\_api\_key\_here
- **Step 6 — Verify the Key is Loaded:** The app.py file loads this key automatically at startup using python-dotenv: from dotenv import load\_dotenv load\_dotenv()  
client = Groq(api\_key=os.environ.get("GROQ\_API\_KEY"))
- **Important — Never Commit Your API Key:** Add .env to your .gitignore file to prevent accidentally pushing the key to a public repository:  
.env

## Activity 1.2: Research and Select the Appropriate Generative AI Model

Here are the things that needed to research to select a best model for the project:



- **Understand the Project Requirements:** Review the specific needs of the CaptionAI application, focusing on the types of content it will generate (captions, hashtags, and calls-to-action across Instagram and LinkedIn).
- **Explore Groq's Model Documentation:** Visit the Groq documentation to examine the available LLM models, including their functionalities, context windows, speed, and limitations.
- **Evaluate Model Performance:** Compare models based on response quality, generation speed, and relevance to social media copywriting tasks.
- **Select the Optimal Model:** Choose **llama-3.3-70b-versatile** for its balance of creative output quality and generation speed, set with a temperature of 0.8 and max\_tokens of 2048 for rich, varied captions.

### Activity 1.3: Define the Architecture of the Application

- **Draft an Architectural Diagram:** Create a visual representation of the application architecture, including components like the frontend (HTML, CSS, JavaScript), Flask backend, and Groq API integration.
- **Detail Frontend Functionality:** Outline how users interact with the application entering product/campaign information, selecting a platform (Instagram or LinkedIn), and choosing a tone (Professional, Casual, or Promotional).
- **Outline Backend Responsibilities:** Specify how the Flask backend handles incoming POST requests to /generate, validates and sanitizes user input, builds a structured prompt, and communicates with the Groq API.
- **Describe AI Integration Points:** Define how the backend constructs prompts using TONE\_DESCRIPTIONS and PLATFORM\_TIPS dictionaries, calls the Groq API, parses the JSON response via extract\_json(), and returns structured results to the frontend.



## Activity 1.4: Set Up the Development Environment

- **Install Python and Pip:** Ensure Python 3.8+ is installed on your machine along with pip for dependency management.
- **Create a Virtual Environment:** Set up a virtual environment using venv to isolate project dependencies:

```
D:\projects>python -m venv myenv  
D:\projects>"d:\projects\venv\Scripts\activate"  
(myenv) D:\projects> pip install -r requirements.txt
```

- **Install Flask:** Use pip to install Flask:

```
(myenv) D:\projects> pip install Flask==3.0.3
```

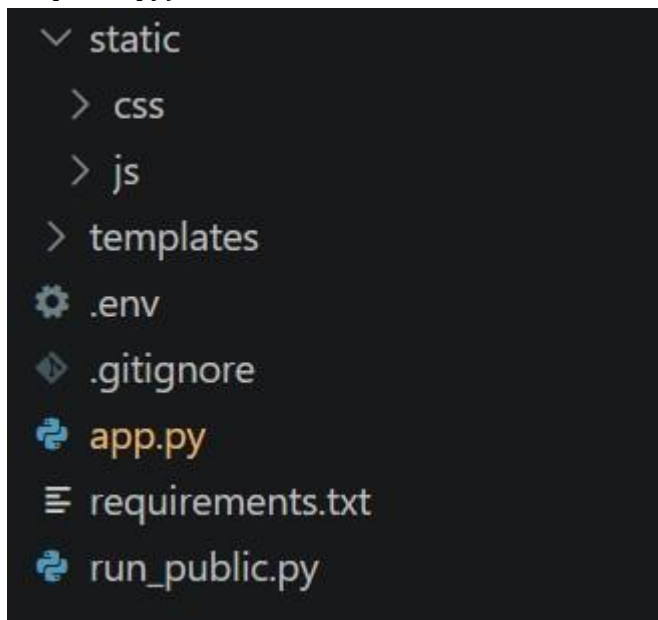
- **Install Groq SDK:** Install the Groq Python library to interact with the Groq API:

```
(myenv) D:\projects> pip install groq==0.11.0
```

- **Install python-dotenv:** Install dotenv support for secure API key management:

```
(myenv) D:\projects> pip install python-dotenv==1.0.1
```

- **Configure the API Key:** Create a .env file in the project root and add your Groq API key:  
GROQ\_API\_KEY=your\_groq\_api\_key\_here
- **Set Up Application Structure:** Create the project directory structure including folders for templates, static files (CSS, JS), and main application files (app.py, requirements.txt, run\_public.py).



## Milestone 2: Core Functionalities Development

## Activity 2.1: Develop Core Content Generation Features

Implement the three core generation features Captions, Hashtags, and CTAs as a unified AI response driven by a single structured prompt.

- **Caption Generator:** Generates three unique, platform-appropriate captions. For Instagram/Casual tone, includes relevant emojis and conversational language. For LinkedIn/Professional tone, produces insight-led, value-driven copy.
- **Hashtag Suggester:** Produces ten hashtags per request a curated mix of high-reach popular tags and targeted niche tags relevant to the product or campaign.
- **CTA Generator:** Returns three short (under 10 words), punchy, action-oriented call-to-action phrases calibrated to the selected platform and tone.

## Activity 2.2: Implement the Flask Backend

### Define Routes in Flask:

- Set up routes in app.py for the home page and content generation endpoint.

### Process User Input:

- Create a form in index.html for users to submit their product/campaign information, platform, and tone.
- Use Flask's `request.get_json()` to retrieve JSON data submitted from the frontend via AJAX.
- Validate that `product_info` is non-empty and within the 2000-character limit before processing.

### Integrate API Calls:

- Within the `/generate` route handler, call the `build_prompt()` function to construct a structured prompt combining tone descriptions and platform-specific tips.
- Submit the prompt to the Groq API using `client.chat.completions.create()` with the `llama-3.3-70b-versatile` model.
- Parse and validate the JSON response using `extract_json()`, which handles direct JSON, markdown code blocks, and raw JSON object extraction as fallback strategies.

## Milestone 3: App.py Development

Milestone 3 focuses on creating and refining the core logic of the `app.py` file, which serves as the backbone of the CaptionAI application. In this milestone, you'll set up each feature's functionality

through Flask routes, establish reliable prompt construction and API interaction, and conduct unit testing to ensure each feature performs as expected.

### Activity 3.1: Writing the Main Application Logic in app.py

#### Define the Core Routes in app.py:

- Establish routes for CaptionAI's two endpoints, each serving a distinct purpose: ◦ / —  
Home page route that renders the main index.html interface

```
@app.route("/")  
  
def index():  
    return render_template("index.html")
```

- **/generate** — POST endpoint that accepts JSON input, invokes the AI, and returns structured content

```
@app.route("/generate", methods=["POST"])  
  
def generate():  
    data = request.get_json()  
    product_info = data.get("product_info", "").strip()  
    platform = data.get("platform", "instagram").lower()  
    tone = data.get("tone", "professional").lower()  
    if not product_info:  
        return jsonify({"error": "Product information is required."}), 400  
    if len(product_info) > 2000:  
        return jsonify({"error": "Input too long. Please keep it under 2000 characters."}), 400  
    try:  
        prompt = build_prompt(product_info, platform, tone)  
  
        chat_completion = client.chat.completions.create(  
            messages=[  
                {  
                    "role": "system",  
                    "content": "You are a social media content expert. Always respond with valid JSON  
only, no markdown formatting."  
                },  
                {  
                    "role": "user",  
                    "content": prompt  
                }  
            ],  
            model="llama-3.3-70b-versatile",  
            temperature=0.8,  
            max_tokens=2048,  
        )  
        response_text = chat_completion.choices[0].message.content
```

```
result = extract_json(response_text)

# Validate structure
if not all(k in result for k in ["captions", "hashtags", "ctas"]):
    raise ValueError("Invalid response structure from AI")

return jsonify({
    "success": True,
    "captions": result["captions"],
    "hashtags": result["hashtags"],
    "ctas": result["ctas"],
    "platform": platform,
    "tone": tone
})

except ValueError as e:
    import traceback
    traceback.print_exc()
    return jsonify({"error": f"Failed to parse AI response: {str(e)}"}), 500

except Exception as e:
    import traceback
    traceback.print_exc()
    return jsonify({"error": f"Generation failed: {str(e)}"}), 500
```

### Set Up Route Handlers for Each Feature:

- For the /generate route, implement logic that reads product\_info, platform, and tone from the incoming JSON body using request.get\_json().
- Apply input validation: return a 400 error if product\_info is empty or exceeds 2000 characters.
- Route AJAX requests from the frontend to the /generate endpoint, which processes data and returns a JSON response.

### Define Prompt Engineering Helpers:

- **TONE\_DESCRIPTIONS** dictionary: maps professional, casual, and promotional tones to detailed copywriting instructions embedded in the prompt.

```
TONE_DESCRIPTIONS = {  
    "professional": "formal, authoritative, and business-oriented. Use clear, concise language that conveys expertise and credibility.",  
    "casual": "friendly, conversational, and relatable. Use informal language, emojis, and a warm tone that feels personal.",  
    "promotional": "exciting, persuasive, and action-driven. Use power words, urgency, and compelling offers to drive engagement."  
}
```

- **PLATFORM\_TIPS** dictionary: maps instagram and linkedin to platform-specific content guidelines (character limits, style expectations).

```
PLATFORM_TIPS = {  
    "instagram": "optimized for Instagram – visually descriptive, emotionally engaging, with strong visual storytelling. Max 2200 characters.",  
    "linkedin": "optimized for LinkedIn – professional insights, value-driven, thought leadership style. Max 3000 characters."  
}
```

- **build\_prompt()**: assembles a structured prompt string from the product info, selected platform tip, and tone description, instructing the model to return strictly valid JSON.

```
def build_prompt(product_info: str, platform: str, tone: str) -> str:  
    tone_desc = TONE_DESCRIPTIONS.get(tone, "engaging")  
    platform_tip = PLATFORM_TIPS.get(platform, "social media")  
  
    return f"""You are an expert social media content strategist and copywriter.
```

**Integrate the Groq AI Responses in Each Function:**

- Call `client.chat.completions.create()` with a system message enforcing JSON-only responses and a user message containing the assembled prompt.
- **extract\_json()**: a robust parser that first attempts `json.loads()`, then strips markdown code fences (````json` blocks), and finally uses a regex to locate a raw JSON object ensuring reliable parsing across model response variations.

```
def extract_json(text: str) -> dict:
    """Extract JSON from LLM response, handling markdown code blocks."""
    # Try direct parse first
    try:
        return json.loads(text)
    except json.JSONDecodeError:
        pass
    # Try extracting from markdown code block
    match = re.search(r'```(?:json)?\s*([\s\S]*?)```', text)
    if match:
        try:
            return json.loads(match.group(1).strip())
        except json.JSONDecodeError:
            pass
    match = re.search(r'\{([\s\S]*?)\}', text)
    if match:
        try:
            return json.loads(match.group(0))
        except json.JSONDecodeError:
            pass
    raise ValueError("Could not extract valid JSON from response")
```

- Validate that the parsed result contains all three required keys — captions, hashtags, and ctas — before returning it to the frontend.
- **Home route:** `/` → renders `index.html/generate` (POST) → accepts JSON, returns `{success,captions, hashtags, ctas, platform, tone}`

## Milestone 4: Frontend Development

Milestone 4 focuses on developing a user-friendly and visually appealing interface for CaptionAI. This involves designing a glassmorphism-style layout with responsive HTML and CSS to enhance usability, and creating dynamic content rendering with vanilla JavaScript to display AI-generated captions, hashtags, and CTAs based on user interactions.

## Activity 4.1: Designing and Developing the User Interface

### Set Up the Base HTML Structure:

- Develop the main HTML file (index.html) as the single-page interface, including a header with the CaptionAI brand logo and tagline, an input section, and a results section.
- Use semantic HTML elements to keep the structure organized and ensure the interface is accessible for all users.
- Integrate Font Awesome icons for platform indicators and result section headings, and Google Fonts (Inter) for clean typography.

### Design a Responsive Layout Using CSS:

- Create a CSS stylesheet (static/css/style.css) implementing a dark glassmorphism aesthetic with animated background blobs (blob-1, blob-2, blob-3) for visual depth.
- Use flexbox and grid layouts to organize the results into a two-column structure (captions on the left, hashtags and CTAs on the right).
- Add media queries to ensure the layout adapts across desktop, tablet, and mobile screen sizes.

### Create Separate UI Components for Each Output Type:

- **Caption Cards:** individual cards per caption, each with a one-click copy-to-clipboard button that provides visual feedback (checkmark + green highlight for 2 seconds).
- **Hashtag Chips:** rendered as pill-shaped chip elements inside a glass card, with automatic # prefix validation.
- **CTA List:** displayed as a clean unordered list inside a separate glass card with a bullhorn icon header.

## Activity 4.2: Creating Dynamic Content Rendering with JavaScript

### Handle Form Submission Asynchronously:

- Attach a submit event listener to the generator form that prevents default form submission and sends the data as a JSON POST request to /generate via the Fetch API.
- During loading, disable the submit button and show an animated CSS loader in place of the button text and arrow icon.

### Render Results Dynamically:



- Implement the `renderResults(data)` function in `main.js`, which dynamically creates and injects DOM elements for captions, hashtags, and CTAs into their respective containers.
- After rendering, automatically smooth-scroll the viewport to the results section using `scrollIntoView()`.

**Restore UI State After Generation:**

- In the `finally` block of the `async` handler, re-enable the submit button and restore the original button text and icon regardless of success or failure.

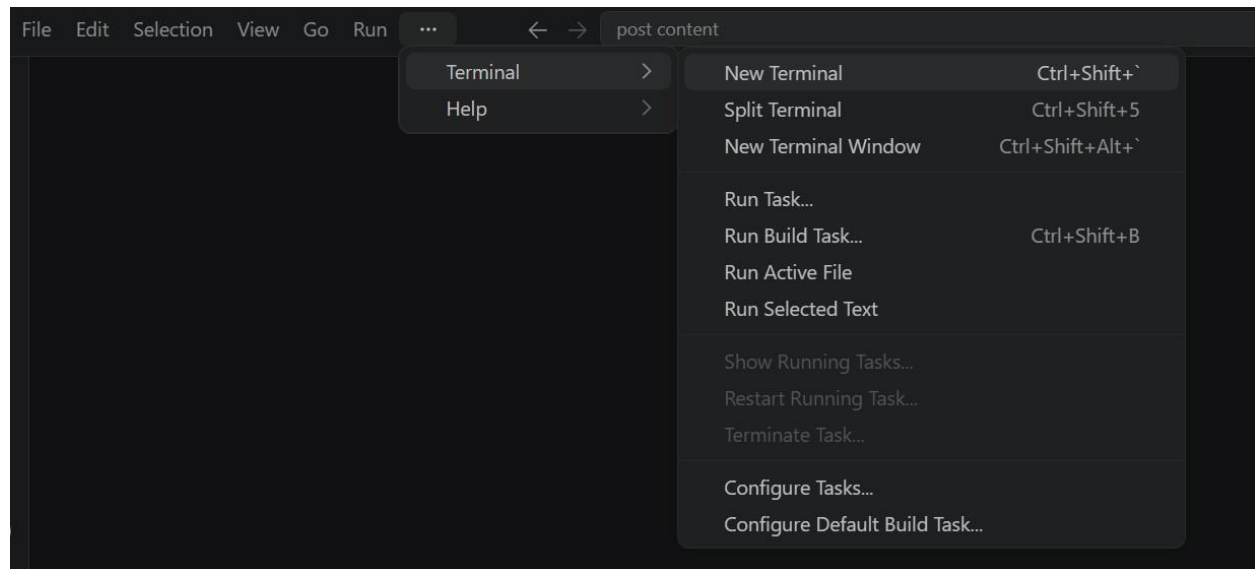
## Milestone 5: Deployment

In Milestone 5, the focus is on deploying the CaptionAI application first locally to verify correct operation, and then publicly via Ngrok to share the app over a secure, accessible URL. This involves configuring the server environment, managing dependencies, and launching the app for real-world use.

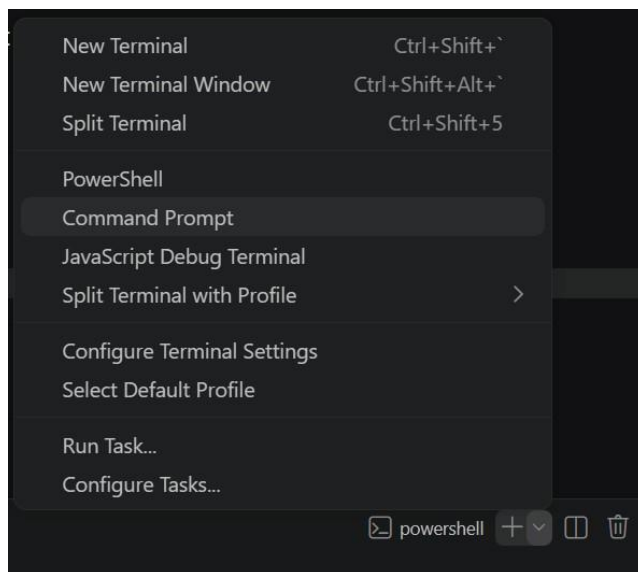
### Activity 5.1: Preparing the Application for Local Deployment

**Set Up a Virtual Environment:**

- Open a New Terminal



- Then open "Command Prompt"



- Create and activate a virtual environment, then install all required dependencies from requirements.

```
D:\projects>python -m venv myenv
D:\projects>"d:\projects\venv\Scripts\activate"
(myenv) D:\projects>pip install -r requirements.txt
```

### Configure Environment Variables:

- Create a .env file in the project root and add your Groq API key. The app loads this automatically using python-dotenv at startup: GROQ\_API\_KEY=your\_groq\_api\_key\_here

## Activity 5.2: Local Testing and Verification

### Start the Flask Development Server:

- Run the application locally using:

```
(myenv) D:\projects>python app.py
```

- Once running, open a browser and navigate to “http://127.0.0.1:5050” to access the app.

```
(venv) D:\projectsSB\AI in healthcare\startup-validator>python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment.
d. Follow link (ctrl + click)
* Running on http://127.0.0.1:5050
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 442-110-195
```

- Interact with the caption generator test all three tones (Professional, Casual, Promotional) and both platforms (Instagram, LinkedIn) to verify the AI responses are correctly generated, parsed, and rendered.

## Activity 5.3: Public Deployment via Ngrok

Ngrok creates a secure tunnel from a public URL to your local Flask server, making the app accessible from anywhere without a cloud hosting provider.

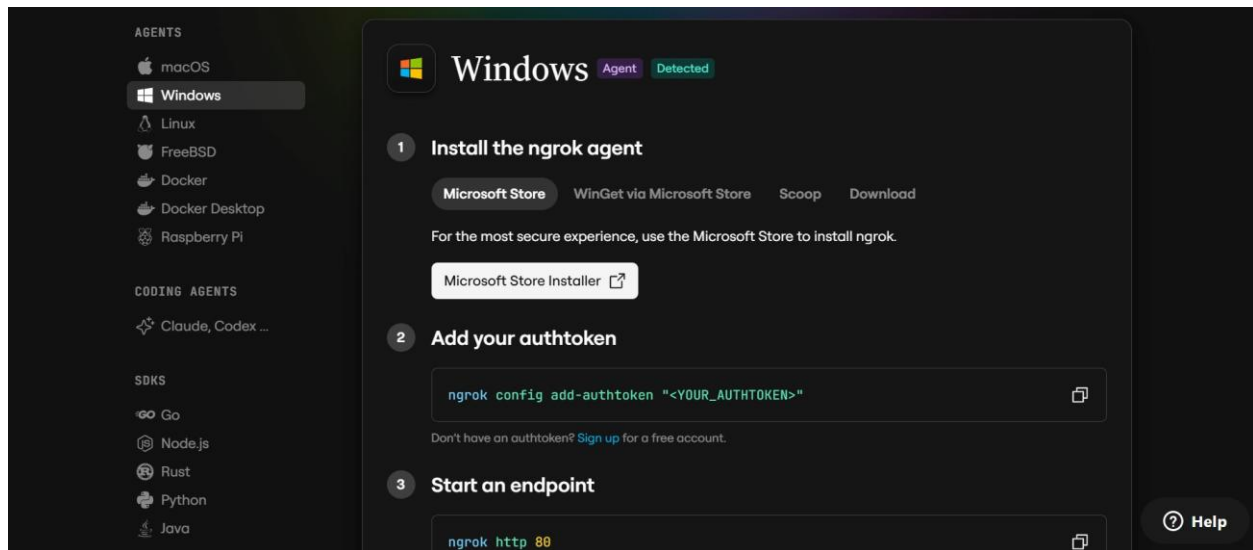
### Install Ngrok and pyngrok:

- Install the pyngrok Python wrapper, which manages Ngrok programmatically within your Python project:

```
D:\projects>pip install pyngrok
```

- Download and install the Ngrok client from <https://ngrok.com/download>.

- Click on “Microsoft Store Installer”.



### Configure Your Ngrok Authtoken:

- Sign up at ngrok.com and copy your authtoken from the Ngrok dashboard.
- The run\_public.py script sets the authtoken automatically using:

```
D:\projects>ngrok.set_auth_token("YOUR_NGROK_AUTHTOKEN")
```

- Replace the TOKEN value in run\_public.py with your own Ngrok authtoken before running.

### Run the Public Deployment Script:

- Instead of running app.py directly, launch run\_public.py to start both the Ngrok tunnel and the Flask server:

```
D:\projects>python run_public.py
```

- The script opens an HTTP tunnel on port 5000 and prints the live public URL in the terminal:  

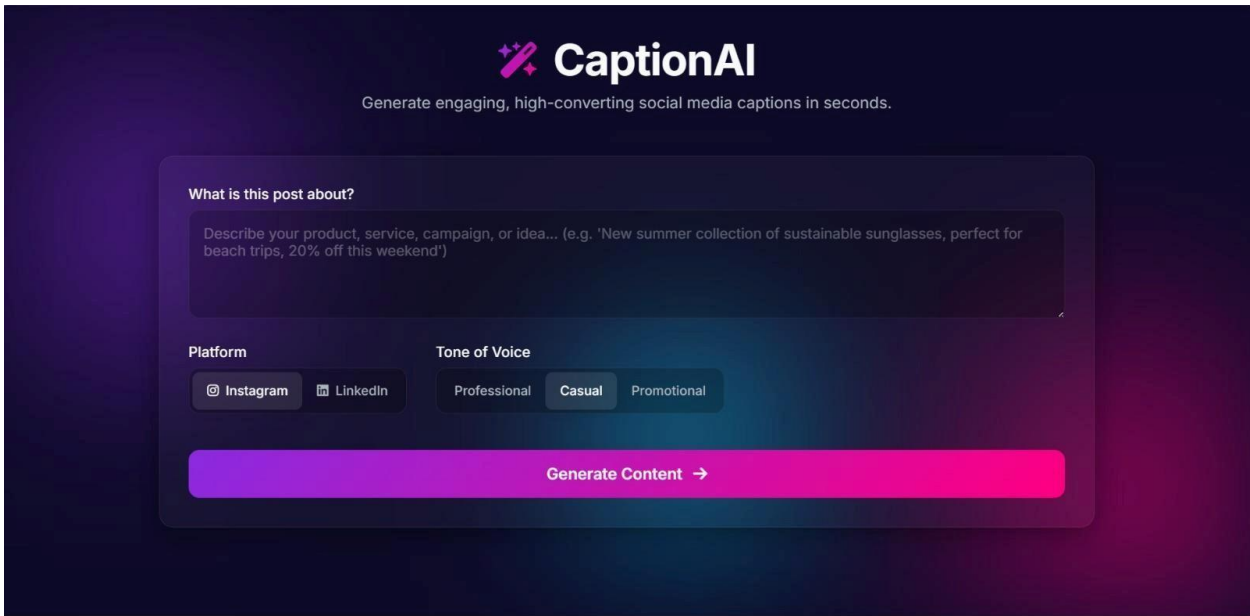
```
===== YOUR
PUBLIC WEBSITE URL IS LIVE!
URL: https://xxxx-xx-xx-xxx-xx.ngrok-free.app
=====
```
- Share this URL with anyone who can access your CaptionAI app directly from their browser without any installation.

### Important Notes:

- **debug=False** is required in run\_public.py to prevent Flask from spawning a reloader process, which would interfere with Ngrok’s tunnel management.
- The public URL changes each time you restart run\_public.py (on the free Ngrok plan). For a persistent URL, upgrade to a paid Ngrok plan and configure a reserved domain.
- Ngrok free tier sessions expire after a few hours of inactivity. Restart run\_public.py to get a new URL when this happens.

## **Exploring the Website's Web Pages:**

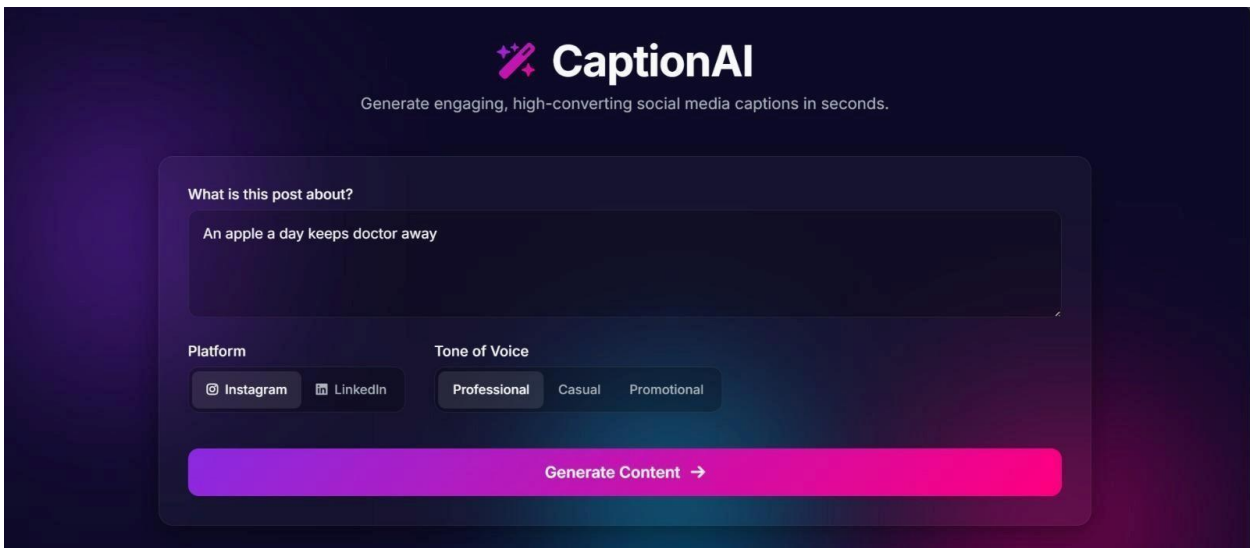
**Home / Generator Page:**



The image shows the CaptionAI web interface. At the top, the brand name 'CaptionAI' is displayed with a wand-magic-sparkles icon. Below it is the tagline 'Generate engaging, high-converting social media captions in seconds.' The main content area is a dark glassmorphism panel. It starts with a text input field titled 'What is this post about?' with a placeholder text: 'Describe your product, service, campaign, or idea... (e.g. 'New summer collection of sustainable sunglasses, perfect for beach trips, 20% off this weekend')'. Below this is a 'Platform' section with two buttons: '@ Instagram' and 'LinkedIn'. To the right is a 'Tone of Voice' section with three buttons: 'Professional', 'Casual', and 'Promotional'. At the bottom of the panel is a large, bright pink button labeled 'Generate Content →'.

**Description:** The single-page interface of CaptionAI serves as both the landing page and the generation tool. It features a dark glassmorphism design with animated background blobs, a header displaying the CaptionAI brand (wand-magic-sparkles icon + logo), and a tagline. The main section contains the input form, platform toggle, tone selector, and the AI-generated results panel all on one seamless page.

## Input Section:



**CaptionAI**  
Generate engaging, high-converting social media captions in seconds.

What is this post about?

An apple a day keeps doctor away

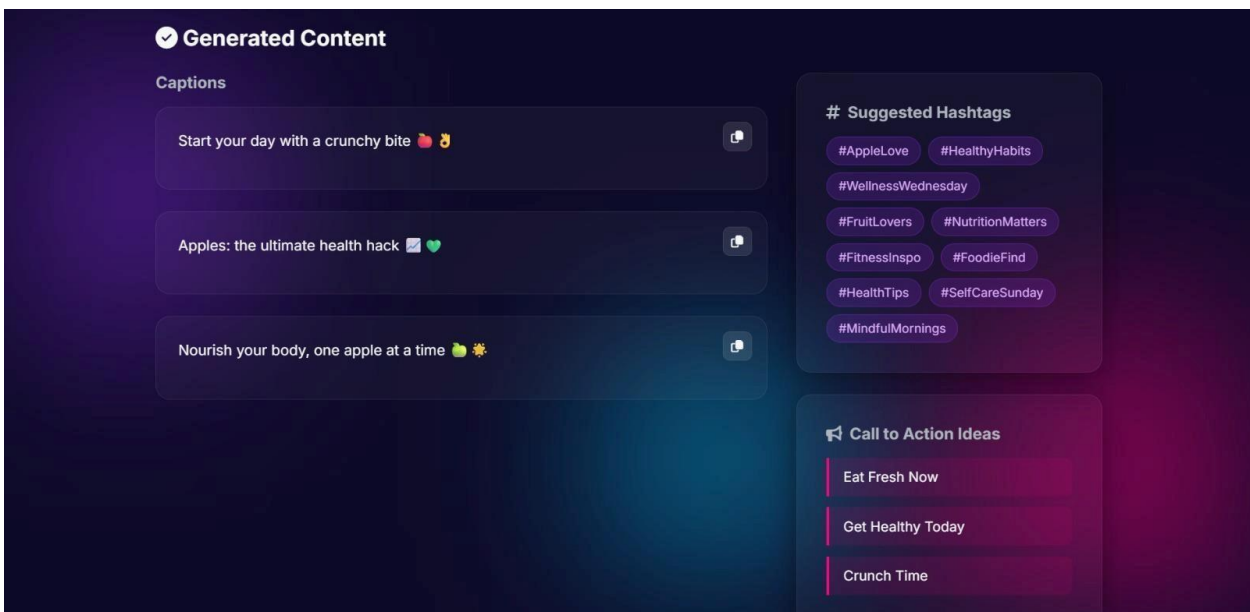
Platform: @ Instagram LinkedIn

Tone of Voice: Professional Casual Promotional

Generate Content →

**Description:** A glass-styled card containing the main generation form. Users enter their product, service, or campaign description in a textarea (up to 2000 characters), select their target platform via a toggle button group (Instagram or LinkedIn), choose their tone of voice (Professional, Casual, or Promotional), and click the Generate Content button. While the AI processes the request, the button transitions to an animated loading spinner.

## Results Section:



✓ Generated Content

Captions

Start your day with a crunchy bite 🍏🔥

Apples: the ultimate health hack 📧❤️

Nourish your body, one apple at a time 🍏☀️

# Suggested Hashtags

#AppleLove #HealthyHabits  
#WellnessWednesday  
#FruitLovers #NutritionMatters  
#FitnessInspo #FoodieFind  
#HealthTips #SelfCareSunday  
#MindfulMornings

🚩 Call to Action Ideas

Eat Fresh Now  
Get Healthy Today  
Crunch Time

**Description:** Revealed dynamically after generation, this section displays the AI output in a two-column grid. The left column shows three caption cards, each with the generated caption text and a copy-to-clipboard button. The right column contains two glass cards: one with ten hashtag chips (prefixed with #) and one with three CTA phrases in a list format. The viewport automatically smooth-scrolls to this section upon content generation.

## Conclusion:

CaptionAI is a generative AI platform built with Flask and styled with a glassmorphism UI that streamlines social media content creation. It uses Groq's high-speed inference API with the LLaMA 3.3-70B Versatile model to instantly generate tailored captions, hashtags, and calls-to-action for Instagram and LinkedIn across professional, casual, and promotional tones. The project, which included model selection, development, and deployment via Ngrok, highlights how targeted AI and a structured framework can boost productivity for marketers and creators, with potential for future scalability. Looking ahead, CaptionAI offers significant opportunities for expansion, such as integrating multi-language support, adding image analysis capabilities for visual-context generation, and implementing user authentication to save generation history. By continuously refining the underlying model prompts and enhancing the frontend accessibility, the platform is well-positioned to evolve from a specialized tool into a comprehensive social media management assistant for teams of all sizes.